

Matías Fernández Lakatos

Senior Data Scientist | Machine Learning Engineer | PhD

Montevideo, Uruguay (Open to US & LATAM Remote)

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PROFESSIONAL SUMMARY

Senior Data Scientist with 9+ years of experience in statistical modeling and machine learning, currently leading the design and validation of unsupervised anomaly detection systems in production cybersecurity environments. Strong expertise in representation learning, high-dimensional modeling, and model evaluation under real-time constraints.

Experienced in end-to-end ML workflows, including experiment tracking (MLflow), hyperparameter optimization, threshold calibration, and integration within distributed streaming systems.

PROFESSIONAL EXPERIENCE

Research Engineer

Gradiant – Spain | 2025 – Present

- Designed and implemented **unsupervised anomaly detection models** (β -VAE, Autoencoders, Isolation Forest variants), **building custom VAE architectures from scratch** using TensorFlow & Keras.
 - Led **training strategy, evaluation framework, and hyperparameter optimization** (grid/random search) using MLflow for experiment tracking, artifact logging, and model versioning.
 - Calibrated anomaly detection thresholds via percentile-based selection on validation data, optimizing decision boundaries with G-mean and loss-based criteria.
 - Evaluated models under **streaming constraints (~300K events)**, balancing detection performance with training/prediction time; resolved memory issues with batch-based processing.
 - Authored **technical documentation for stakeholders (INCIBE)** and collaborated with engineering/security teams in model integration and peer code reviews.
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Researcher & Teaching Assistant

Universidad de la República – Uruguay | 2015 – 2024

- Led independent computational research projects in statistical modeling and signal processing.
 - Designed numerical methods and data-processing pipelines for high-dimensional systems.
 - Published several peer-reviewed research in international journals.
 - Supervised undergraduate and graduate students in quantitative modeling.
 - Delivered university-level courses in physics, mathematics, and computational methods.
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EDUCATION

MSc, Big Data Analytics – Universidade de Santiago de Compostela (2025)

PhD, Optics – Universidad de la República (2024)

MSc, Theoretical Physics – Universidad de la República (2018)

TECHNICAL SKILLS

Programming & Data: Python (pandas, numpy), SQL, R | **Machine Learning & Deep Learning:** Unsupervised Learning, Representation Learning, beta Variational Autoencoders (b-VAE), Deep Neural Networks, TensorFlow & Keras (custom architectures), Scikit-learn, Hyperparameter Optimization, Model Validation, Threshold Calibration | **Tools & Pipelines:** MLflow | **Systems & Infrastructure:** Distributed ML environments, Streaming architectures, Linux | **Other:** Git, Digital Image Processing